GBDA 412: AUDIO CULTURES

Musical Algorithms as Tools, Languages & Partners

Alex McLean & Roger T. Dean (2018) — The Oxford Handbook of Algorithmic Music, Ch. 1

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s("bd sd [~bd] sd, $hh*8")
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WARM-UP HANDS UP

Which of these is an algorithm?

A

A recipe

"Cream butter and sugar. Add eggs one at a time. Bake for 25 minutes."

MANUAL STEP

B

A 1960s score

"A source of dripping water and an empty vessel..." — George Brecht, Drip Music

PROCESS MUSIC

C

A prompt

"Upbeat lo-fi track about cramming for midterms, soft vocals." Typed into AI.

GENERATIVE PROMPT

All three! The reading defines an algorithm as *"a finite sequence or structure of instructions."*

CORE SYNTHESIS

Four Main Arguments of the Text

01

Old Idea, New Conditions

Composing with rules is centuries old; cheap, open computers are what's new.

02

Tool, Language or Partner

A spectrum of agency stretching from human notation to independent co-performers.

03

No Turing Test Needed

Judge music by its novel "meta-human" outputs, not by whether it tricks human ears.

04

Open Access & Open Canon

Code must be open to all, acknowledging historical gaps in gender and global representation.

ARGUMENT 02

Tools, Languages, Partners: Who Holds Agency?

HUMAN AGENCY

Tool / Notation

The human writes and rewrites the rules. Live coding: the musician edits code mid-performance.



IN BETWEEN

Language

Code as a way to think and express musical ideas, moving closer to written natural languages.



MACHINE AGENCY

Partner

An independent agent that listens and responds. Live algorithms act as non-human co-performers.



CLOSING TAKEAWAY

The algorithm is centuries old. The real question is who holds agency: tool, language, or partner?

Thank you — Min Htet Naing & Nick Wang

ANALYTICAL FRAMEWORK

The Four Pillars of the Text

ARGUMENT 1

Tools, Languages & Partners

Reframing computational agency across an interactive continuum.

ARGUMENT 2

Meta-Human Creativity

Moving past style imitation toward novel perceptual frontiers.

ARGUMENT 3

Live Improvisation

Real-time execution via live coding and machine listening.

ARGUMENT 4

Cultural Assimilation

Computational literacy democratizing music practice.

ARGUMENT 1: AGENCY SPECTRUM

Algorithms as Tools, Languages, and Partners

1. Algorithm as Tool

Automates routine compositional tasks, score notation processing, and sound spatialization.

2. Algorithm as Language

Serves as a dynamic medium for human expression and formal musical thought construction.

3. Algorithm as Partner

Acts as an autonomous co-performer with machine listening and generative response capabilities.

ARGUMENT 2: BEYOND IMITATION

Rejecting Imitation for Meta-Human Possibilities

Beyond the Turing Test

Evaluating algorithmic music by its ability to mimic past human styles restricts its creative potential.

Meta-Human Horizons

Algorithms uncover complex acoustic and structural forms unachievable through manual unassisted human composition.



ARGUMENT 3: REAL-TIME PRACTICE

Improvisation & Live Execution



From Offline to Real Time

Historical CPU limits bound algorithms to offline studio composition; modern chips enable live performance.

Live Coding & Transparency

Projecting code turns the process into a public visual score, breaking performance determinism.

Machine Listening Systems

Live algorithms analyze acoustic input, permitting adaptive human-machine conversational dialogue.

ARGUMENT 4: CULTURAL INTEGRATION

Democratization & Cultural Assimilation

Ubiquitous Computation

Low-cost hardware (Raspberry Pi) and open-source software move practice out of academic labs.

Pedagogical Shift

Environments like Sonic Pi integrate programming directly into basic creative music education.

Everyday Human Expression

Subcultures (Algoraves, Minecraft music) re-humanize automation into shared social art forms.

CONCLUSION

Re-Humanizing Automation Through Formal Art

McLean & Dean demonstrate that algorithmic music is not a replacement for human creativity, but an expanded language for thinking, improvising, and sharing music in a computational age.

Image Sources



http://kraftmusic.com/cdn/shop/files/ASM-LSDRCK-2026-07-31-patch-cable-color-swap.jpg?v=1785524477&width=3000

Source: kraftmusic.com



https://media.redbullmusicacademy.com/assets/601/623/274/algorave-title.jpg?w=1080

Source: daily.redbullmusicacademy.com